
MLPC Report - Task 4: Challenge

Team OBSERVE

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Contributions

All of us together were responsible for tasks 1.) and 2.) . All of us together were responsible for tasks 3.) and 4.) . All of us together were responsible for task 5.) Bonus. In the same constellation we created this report. We all worked on the presentation together, with Group 2 providing its content. We held regular meetings, where each group presented their results up to that point and the other critically reviewing their work. Collaboration was done over GitHub and project progression was tracked through regular stand-ups.

1.) Describe your evaluation setup

- a.) How did you split the development dataset (provided in phase three of the project) into a training and evaluation set while avoiding information leakage?
- b.) Establish and describe a naive baseline system (i.e., a baseline without a classifier). How much cost can we expect from it?

2.) Build a simple SED system, that predicts the presence and absence of the ten sound categories in 1.2-second audio segments.

- a.) How did you threshold and combine predictions to derive a label for the 1.2-second segments? Describe any heuristics or post-processing strategies used.
- b.) What strategies did you apply to minimize the task-specific cost function?
- c.) Does your classifier-based SED system achieve lower costs compared to the naive baseline system you established in step 1? If not, describe possible issues.

3.) Investigate at least three diverse strategies to improve your starting point

- a.) For each of these strategies: Describe the main ideas you had and hypotheses you tried, as well as their outcome after experimentally verifying or falsifying the hypotheses (via validation on (parts of) the development set). Do not hesitate to report negative outcomes.

4.) Do you think your final system could be deployed in a real-world application?

- a.) In your opinion, which aspects of the project or your system would need to be adapted to fulfill possible real-world requirements?

5.) Bonus

So far, we've worked with audio embeddings, that were created from a pre-trained and frozen sound event detection model. To receive the bonus points, train one of our pre-trained sound event detection models on the MLPC dataset to

predict the ten sound categories of interest in an end-to-end manner. You can find an implementation, a detailed documentation and pre-trained checkpoints here: <https://github.com/fschmid56/PretrainedSED>. Use one of the checkpoint that was pre-trained on AudioSet-strong as a starting point. You can use any of the model architectures. Compare the performance of the resulting system to the best model from the previous section.

- a.) Train one of our pre-trained sound event detection models on the MLPC dataset to predict the ten sound categories of interest in an end-to-end manner.**
- b.) You can use any of the model architectures.**
- c.) Compare the performance of the resulting system to the best model from the previous section.**